

WIRELESS CHARGING: Fuel Of The Future

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Wireless charging will increase mobility for users of Internet of Things (IoT) devices. This article explores the technology and its applications, challenges to overcome while designing and the latest advances in this field

Wireless charging eliminates the requirement of charging cables and allows charging of multiple devices in parallel. There are no charging cable incompatibility issues. Basically, wireless charging works on the principle of electromagnetic induction: Time-varying magnetic field induces current in a closed loop of wire to transfer power from the transmitter to the receiver without requiring physical connection.

Wireless charging allows auto shut-down when charging is complete. As it does not require mechanical connectors, manufacturing cost is less. Embedding this technology in mobiles will allow mobile manufacturers to do away with power supply sockets, which allow entry to water, dust and other corrosive materials.

Application areas

Apple announced its wireless charging mat AirPower, which is capable of charging iPhone, Apple Watch and AirPods simultaneously. Ikea provides a wide range of wireless charging furniture that support Qi standard. Wireless charging-compatible devices include notebooks, tablets, smartphones, wearables, multicopters, electric toys, cameras, speakers, power tools, service robots such as vacuum cleaners, in-car chargers, electric vehicles, medical devices, etc. Wireless charging can also be used to recharge embedded medical devices via a magnetic field through the skin, avoiding the risk of infection.

Mediatek principal engineer Sandeep Arya informs that their company provides

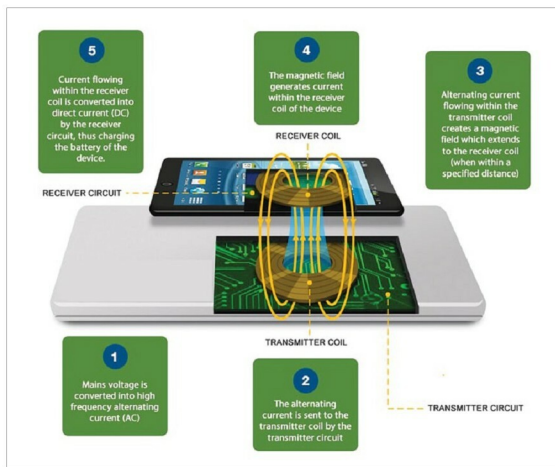


Fig. 1: Work flow of a wireless charger (Image source: <http://electronicstrick.blogspot.in>)